Magpie — Sprint 17 spec: photo fix-up (local, own photos only) — FINAL SPRINT

Audience: Codex. **Author:** Claude. **Status: ready to implement. Locks:** D-124 (photo policy), D-127 (this sprint's shape). **References:** Sprint 1 (the photo upload pipeline + EXIF stripping this builds on), the per-field Approve/Edit/Reject review spine (Sprint 13), F-14 (Windows/SQLite/Waitress runtime). **This is the last sprint — closing it completes Phase 5 and the entire original roadmap.**

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1. Goal

One-tap, all-local clean-up of the user's OWN photos, proposed for human approval and applied non-destructively. One tap runs the whole pipeline — crop, straighten/rotate, white-balance, exposure, background clean-up — and shows a **before/after**; the user **approves, tweaks, or rejects**. The original is always kept. **Zero cloud** (runs on the box). It makes *the user's own* photos look clean and true — **never** altering how the item's condition reads, **never** importing anyone else's photos.

2. Scope

IN (v1):

- **One-tap fix-up** running the full **local** pipeline at once (§4).
- **Look-and-approve** review — before/after, Approve / Tweak / Reject (§5).
- **Non-destructive** — fix-up is a new version; the original is always retained + revertable (§6).
- **Batch-friendly review** — process an item's photos and thumb-through-approve in one pass (§7).

OUT (explicitly):

- ❌ **No cloud / API / network** — entirely local this sprint. *(Cloud enhancement is **deferred as a possible future optional extra** — D-127 — and is not built here. If ever added, it sits behind the same approve step + condition-integrity guards.)*
- ❌ **No auto-apply** — the fix-up is **user-triggered and human-approved**; it never

silently rewrites a photo on upload.

- ❌ **No destructive edits** — the original is never overwritten/lost.
- ❌ **Nothing that alters how condition reads** — no smoothing/removing scratches or wear, no boosting shine/gloss, **no generative or "beautify"/retouch step** (D-124). Operations are limited to framing, orientation, lighting, and background.
- ❌ **No third-party / "professional" / stock / other-seller images** — only the user's own photos (D-124).

3. Data model (additive)

- Photos already exist (Sprint 1 upload pipeline, EXIF-stripped). The fix-up produces a **derived "fixed" version linked to the original**, with the **original always retained**. Add a derived-image record (or a version relationship) + a per-photo fix-up **state** (`none` / `pending-review` / `approved` / `rejected`). The **approved fixed version becomes the display image**; the original stays revertable.
- Additive migration only; **pre-migration encrypted backup + restore spot-check** (new derived-image storage).

4. The local pipeline (deterministic, conservative)

One tap runs all of these **on the box, offline**:

- **Straighten / rotate** (deskew; respect existing EXIF orientation handling).
- **Auto-crop** to sensible framing of the item (don't clip the item).
- **White-balance** (e.g. gray-world / auto) — true colour, *gently*.
- **Exposure / contrast** (auto-levels, gentle).
- **Background clean-up / removal** → a **clean (white) background**, via a **local offline model** (e.g. an `rembg` /U²-Net-style segmenter run via onnxruntime — **bundled, no network**). **Verify it runs in the Windows / NSSM / Waitress runtime (F-14)**; if a local model genuinely can't run offline there, fall back to a simpler background clean-up (threshold/flood-fill on near-uniform backgrounds) — but try the model first (Regan chose the full set).
- **Tuning is conservative:** the pipeline must **not distort the item or how its condition reads**. When uncertain (e.g. a risky background mask near the item edge, an aggressive balance shift), **prefer under-processing**. There is **no retouching/generative step** — this is geometry + levels + background only.

5. Look-and-approve review

- The fix-up shows a **before/after** of the result; the user **Approves** (commit the fixed version as the display image), **Tweaks** (basic manual adjust — e.g. re-crop/rotate — then approve), or **Rejects** (discard, keep original).
- **Nothing is applied without an explicit approve** — same human-in-the-loop guarantee as the rest of Magpie (test-prove it: a generated fix-up does **not** become the display image until approved; reject leaves the original as the display image).

6. Non-destructive

- The **original is always retained** and the photo can be **reverted to original** at any time. The fixed version is a derivative, not a replacement. (No path destroys the original.)

7. Batch-friendly review

- For an item with several photos, the user can **run fix-up across them and review/approve in one pass** (thumb through before/afters), rather than one-photo-at-a-time ceremony. Per-photo approve/reject still applies — batch is about *review flow*, not auto-approval.

8. Design

Reuse the **bright, high-contrast** tokens from Sprint 16 (legible on a phone). The before/after is clear; Approve/Tweak/Reject are obvious finger-targets. Responsive; reduced motion. Use the `frontend-design` skill for conventions.

9. Acceptance criteria (closure gate)

- **One tap** runs the full local pipeline (crop/straighten/rotate/white-balance/exposure/background) and shows a **before/after**.
- **Approve** commits the fixed version as the display image; **the original is retained + revertable**; **Reject** keeps the original; **nothing is applied without approval** (test-proven).
- **Batch:** an item's multiple photos can be processed + reviewed/approved in one pass.
- **All-local:** works with **no cloud / no network / no API** — assert no external calls; the

background model runs **offline**.

- **Condition integrity:** the pipeline has **no generative/retouch/beautify step**; operations limited to framing/orientation/lighting/background (assert in code + a manual before/after check in evidence showing wear/scratches are **not** smoothed away).
- **No third-party images** anywhere in the flow.
- **Process:** backend + frontend tests green; **dual-lane Validation green**; rebuild + restart the `Magpie` service; additive migration; **pre-migration encrypted backup + restore spot-check**; **before/after screenshots** (desktop + phone) are the headline evidence. Evidence as `SPRINT17_EVIDENCE.md`.

10. For Codex

Reasoning: High. Entirely local — no cloud/network/API this sprint (cloud enhancement is deferred, D-127). **User-triggered + human-approved — never auto-apply** on upload. **Non-destructive — never overwrite/lose the original.** Background removal via a **local offline model verified to run in the Windows/Waitress runtime** (offline; fall back to simpler background clean-up only if a local model truly can't run there). **No generative/retouch/beautify step and nothing that changes how condition reads** (D-124) — geometry + levels + background only; prefer under-processing when uncertain. **No third-party images.** Additive migration; backup + restore before touching the runtime. Don't print/commit secrets/keys/`.env`/tokens/passphrases. Evidence as `SPRINT17_EVIDENCE.md`. **This is the final sprint — closing it completes Phase 5 and the entire original roadmap.**